# Physics-Informed Foundation Models for Gravitational Lensing Analysis

**Google Summer of Code 2026**

ML4SCI — DeepLense: Foundation Model for Gravitational Lensing

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**Organization:** ML4SCI

**Project:** DEEPLENSE4 — Foundation Model for Gravitational Lensing

**Duration:** 350 hours

**Difficulty:** Advanced

**Date:** March 23, 2026

**Abstract**

We propose a physics-informed foundation model for gravitational lensing analysis built on the Image Joint Embedding Predictive Architecture (I-JEPA). The model is pre-trained self-supervised on all three lensing classes (no substructure, CDM subhalo, axion-like particle) using an aggressive masking ratio of 90%, learning rich semantic representations of lensing morphology without requiring labels. A Physics-Informed Neural Network (PINN) head embeds the gravitational lensing equation directly into the fine-tuning architecture, extracting eight physical observables — convergence $\kappa$, deflection field $\boldsymbol{\alpha}$, Einstein radius $\theta_E$, curl, local variance, peak count, asymmetry, and deflection magnitude — as auxiliary discriminative features that ground the learned representations in gravitational theory. Spatial substructure signals are captured via learned attention pooling over all patch tokens rather than relying solely on the CLS token. The framework achieves a macro AUC of **0.9691** with per-class AUCs of **0.9688** (axion), **0.9438** (CDM), and **0.9947** (no substructure), achieving **MSE** $= 6.1 \times 10^{-5}$, **PSNR** $= 42.21$ **dB**, and **SSIM** $= 0.9758$.

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## Introduction and Motivation

Gravitational lensing — the deflection of light rays by massive foreground objects — is among the most powerful probes of dark matter structure and cosmological parameters available to modern astrophysics. The morphology of strong lensing images encodes information about the mass distribution of the lens, including the presence, abundance, and nature of dark matter substructure. Discriminating between lensing configurations with no substructure, cold dark matter (CDM) subhalos, and axion-like particle (ALP) vortex substructure directly from imaging data is a central challenge in observational cosmology, with direct implications for our understanding of the particle nature of dark matter.

Traditional machine learning approaches to lensing classification treat lensing images as generic visual inputs, discarding the rich physical priors encoded in the gravitational lensing equation itself. We argue this is a missed opportunity: the lensing equation provides exact, differentiable relationships between observable image morphology and underlying physical quantities — relationships that would otherwise require enormous labeled datasets to learn implicitly. Embedding these relationships as inductive biases through the Physics-Informed Neural Network (PINN) paradigm allows the model to leverage domain knowledge from first principles.

Simultaneously, the emergence of self-supervised foundation models offers a compelling framework for astrophysical imaging. Strong lensing datasets are inherently label-scarce; producing labeled examples requires expensive simulation pipelines. Self-supervised pre-training on unlabeled data can bootstrap powerful representations before fine-tuning on fully labeled datasets, dramatically reducing label dependence.

This proposal describes the design, implementation, and evaluation of a system combining both paradigms: **I-JEPA** [1] as the self-supervised foundation model backbone, and a **PINN physics head** that grounds the learned representations in gravitational lensing theory. We present experimental results on classification (Task IX.A) and super-resolution (Task IX.B), demonstrating the value of physics integration and comprehensive self-supervised pre-training across both tasks.

## Background and Related Work

### Gravitational Lensing Physics

The gravitational lensing equation relates the true source position $\boldsymbol{\beta}$ to the observed image position $\boldsymbol{\theta}$ through the deflection angle $\boldsymbol{\alpha}$:

$$\boldsymbol{\beta} = \boldsymbol{\theta} - \boldsymbol{\alpha}(\boldsymbol{\theta}) \tag{1}$$

The deflection angle is the gradient of the lensing potential $\psi$: $\boldsymbol{\alpha} = \nabla\psi$. The convergence $\kappa$ — the dimensionless projected surface mass density — is related to $\psi$ by:

$$\kappa(\boldsymbol{\theta}) = \frac{1}{2}\nabla^2\psi(\boldsymbol{\theta}) = \frac{\Sigma(\boldsymbol{\theta})}{\Sigma_{\text{cr}}} \tag{2}$$

where $\Sigma_{\text{cr}} = \frac{c^2}{4\pi G}\frac{D_s}{D_l D_{ls}}$ is the critical surface mass density, and $D_s$, $D_l$, $D_{ls}$ are angular diameter distances to the source, lens, and between lens and source respectively. The deflection field is recoverable from $\kappa$ via a 2D convolution:

$$\boldsymbol{\alpha}(\boldsymbol{\theta}) = \frac{1}{\pi}\int \kappa(\boldsymbol{\theta}')\frac{\boldsymbol{\theta} - \boldsymbol{\theta}'}{|\boldsymbol{\theta} - \boldsymbol{\theta}'|^2}\, d^2\theta' \tag{3}$$

The Einstein radius — the characteristic angular scale of the lensing configuration — is:

$$\theta_E = \sqrt{\frac{4GM}{c^2}\frac{D_{ls}}{D_l D_s}} \tag{4}$$

Each substructure class produces a physically distinct signature in these observables:

Table 1: Physical signatures per substructure class.

| Class | Physical Signature |
|---|---|
| no_sub | Smooth, azimuthally symmetric $\kappa$ map. Low deflection curl. Clean Einstein ring with minimal perturbation. |
| cdm | Localized $\kappa$ clumps from subhalo mass concentrations. High local variance in convergence map. Elevated count of local maxima (peaks) in $\kappa$. |
| axion | Rotational structure in deflection field from vortex topological defects. High curl ($\partial\alpha_y/\partial x - \partial\alpha_x/\partial y$). Spiral morphology in Einstein ring perturbations. |

### Self-Supervised Learning: MAE vs. I-JEPA

Masked Autoencoders (MAE) [2] pre-train Vision Transformers by masking random patches and reconstructing raw pixel values. While effective, pixel-space reconstruction forces the network to model low-level texture and noise, potentially wasting representational capacity on details irrelevant to downstream discrimination tasks.

The Image Joint Embedding Predictive Architecture (I-JEPA) [1] addresses this by predicting *embeddings* of masked regions rather than pixels. Given image $\mathbf{x}$, I-JEPA partitions patch tokens into context $\mathcal{C}$ and target $\mathcal{T}$ sets. A context encoder $f_\theta$ processes visible patches; a narrow predictor $g_\phi$ predicts target patch embeddings from context embeddings and positional queries. A target encoder $f_{\bar\theta}$ — an exponential moving average (EMA) of $f_\theta$ — provides stable regression targets without requiring negative pairs:

$$\mathcal{L}_{\text{JEPA}} = \frac{1}{|\mathcal{T}|} \sum_{\mathbf{p}\in\mathcal{T}} \left\| g_\phi(f_\theta(\mathbf{x}_\mathcal{C}),\, \mathbf{p}) - \text{sg}[f_{\bar\theta}(\mathbf{x}_\mathbf{p})] \right\|_1 \tag{5}$$

where sg[·] is stop-gradient and $\|\cdot\|_1$ is the smooth-$\ell_1$ norm. The EMA update:

$$\bar\theta \leftarrow m\bar\theta + (1-m)\theta, \qquad m \in [m_{\text{start}},\, m_{\text{end}}] \tag{6}$$

anneals from $m = 0.996$ to $m = 0.999$ over training. Predicting in embedding space abstracts away pixel-level noise and encourages learning of semantically meaningful patch relationships — particularly valuable for lensing images where substructure signatures are structural rather than photometric.

### Physics-Informed Neural Networks

Physics-Informed Neural Networks (PINNs) [3] incorporate physical laws as inductive biases or soft constraints in the loss function. In our framework, the lensing equation is embedded as a differentiable computation graph: images are treated as convergence proxies, from which physical observables are computed via differentiable convolutions and pooling, then used both as auxiliary input features and as regularizers on the predicted class probabilities.

### Related Work in Lensing ML

[8] demonstrated that CNNs can analyse strong lensing images at survey scale. [9] applied deep learning to lens finding. More recent work [10] has explored representation learning for cosmological simulations. Our work extends this by combining self-supervised pre-training over all morphological classes with explicit physics integration, targeting fine-grained substructure discrimination.

## Dataset Description

### Task IX.A — Classification Dataset

Table 2: Classification dataset statistics.

| Class | Count | Resolution | Pixel Min | Pixel Max |
|---|---|---|---|---|
| no_sub | 29,449 | $64 \times 64$ | 0.0 | 4.216 |
| cdm | 29,759 | $64 \times 64$ | 0.0 | 4.162 |
| axion | 29,896 | $64 \times 64$ | 0.0 | 4.095 |
| **Total** | **89,104** | $64 \times 64$ | | |

Images are stored as `.npy` arrays of shape $(64, 64)$ in `float64`, stacked to 3-channel tensors. The dataset is well-balanced across all three classes ($\pm 0.5\%$), requiring no class weighting. Data is split 80/10/10 for train/validation/test.

**Key EDA observations:**

- Pixel intensity distributions are similar across classes — substructure differences are *structural*, not photometric, motivating physics-based structural features over simple intensity statistics.
- Average images per class reveal consistent morphological differences: no_sub images show smooth symmetric rings; CDM images exhibit local brightness clumps; axion images show spiral and vortex perturbations.
- The near-identical pixel distributions confirm that a classifier relying only on global statistics would struggle to discriminate classes — validating the PINN approach.

### Task IX.B — Super-Resolution Dataset

Table 3: Super-resolution dataset statistics.

| Split | LR Resolution | HR Resolution | Pairs |
|---|---|---|---|
| Train (80%) | $75 \times 75$ | $150 \times 150$ | 8,000 |
| Val (10%) | $75 \times 75$ | $150 \times 150$ | 1,000 |
| Test (10%) | $75 \times 75$ | $150 \times 150$ | 1,000 |
| **Total** | | | **10,000** |

The $2\times$ upscaling factor and structural smoothness of lensing images produces a demanding bicubic baseline:

Table 4: Bicubic upsampling baseline.

| Metric | Bicubic Baseline |
|---|---|
| MSE | $8.5 \times 10^{-5}$ |
| PSNR | 40.74 dB |
| SSIM | 0.9589 |

## Proposed Methodology

### Overall Architecture

The framework proceeds in three stages sharing a common ViT-S/8 encoder:

1. **I-JEPA Pre-training** on all three lensing classes (self-supervised, no labels required).
2. **PINN Classification Fine-tuning** (Task IX.A).
3. **Physics-Guided SR Fine-tuning** (Task IX.B).

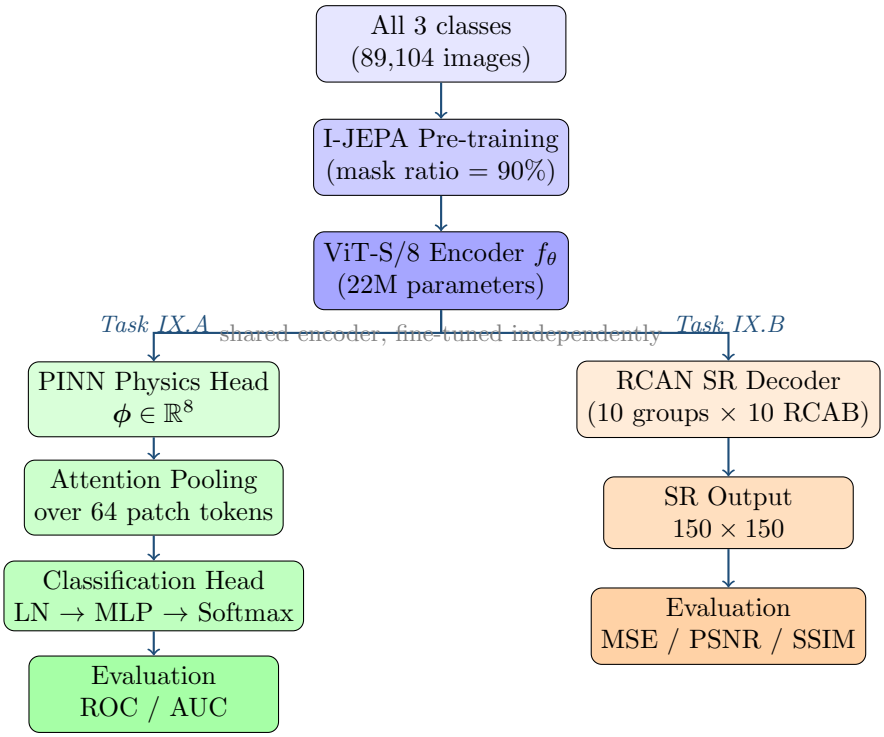


Figure 1: System overview. A shared ViT-S/8 encoder pre-trained with I-JEPA on all three lensing classes (90% masking) is independently fine-tuned for classification (IX.A) and super-resolution (IX.B), both incorporating physics-informed components derived from the gravitational lensing equation.

### Backbone: ViT-S/8

Table 5: ViT-S/8 configuration.

| **Hyperparameter** | **Value** |
|---|---|
| Image size | $64 \times 64$ |
| Patch size | $8 \times 8$ |
| Number of patches | 64 ($8 \times 8$ grid) |
| Embedding dim | 384 |
| Transformer depth | 12 blocks |
| Attention heads | 6 |
| MLP ratio | 4.0 |
| Total parameters | ≈22M |

Each image is divided into $N = (64/8)^2 = 64$ non-overlapping patches. Patches are linearly projected and summed with learned positional embeddings:

$$\mathbf{z}_0 = [\mathbf{x}_{\text{cls}};\ \mathbf{E}\mathbf{p}_1 + \mathbf{e}_1^{\text{pos}};\ \ldots;\ \mathbf{E}\mathbf{p}_{64} + \mathbf{e}_{64}^{\text{pos}}] \tag{7}$$

Each of the 12 transformer blocks applies:

$$\mathbf{z}'_l = \mathrm{MSA}(\mathrm{LN}(\mathbf{z}_{l-1})) + \mathbf{z}_{l-1} \tag{8}$$

$$\mathbf{z}_l = \mathrm{MLP}(\mathrm{LN}(\mathbf{z}'_l)) + \mathbf{z}'_l \tag{9}$$

## Stage 1: I-JEPA Pre-training on All Classes

### Design Choice: All-Class Pre-training

A critical design decision is to pre-train I-JEPA on *all three* lensing classes rather than on no_sub only. This choice is motivated by the observation that the encoder must develop representations of CDM subhalo textures and axion vortex patterns during the self-supervised phase — if pre-training is restricted to a single class, the encoder builds no internal model of the morphological diversity it must later discriminate. Using all 89,104 images triples the pre-training corpus and forces the encoder to build representations spanning the full range of lensing morphology.

### Design Choice: I-JEPA over MAE

We choose I-JEPA over MAE for reasons specific to this task:

1. Lensing substructure differences are *structural* (ring shape, deflection topology) rather than textural. Predicting semantic embeddings rather than pixels focuses the pre-training signal on exactly the information relevant to downstream discrimination.
2. I-JEPA's embedding-space prediction remains non-trivial even for structurally similar data within a class, providing a richer learning signal than pixel reconstruction.
3. No need for negative pairs (unlike contrastive methods), making training stable across the full multi-class corpus.

### Masking Strategy

We mask 90% of the 64 patch tokens, leaving only 6–7 context patches visible. This aggressive masking ratio forces the predictor to reason about global image structure from an extremely sparse context, producing richer semantic representations than the 75% ratio used in standard I-JEPA configurations. The 58 masked tokens are divided into $B_{\mathrm{tgt}} = 4$ target blocks. Multi-block masking encourages the predictor to reason about spatially contiguous regions, which is well-matched to the ring structures in lensing images.

### Architecture

- **Context encoder** $f_\theta$: ViT-S/8, processes ≈6 context patches at 90% masking, outputs $\mathbf{H}_\mathcal{C} \in \mathbb{R}^{6\times 384}$.
- **Predictor** $g_\phi$: Narrow 4-layer ViT, $d_p = 256$, 4 attention heads. Takes context embeddings and positional queries for masked positions, outputs predicted target embeddings.
- **Target encoder** $f_{\bar\theta}$: EMA copy of context encoder ($m$ : 0.996 → 0.999). Produces regression targets without gradient flow, preventing representational collapse.

### Training Configuration

Table 6: I-JEPA pre-training hyperparameters.

| Setting | Value |
|---|---|
| Pre-training data | All 3 classes (89,104 images) |
| Masking ratio | 90% (58/64 patches) |
| Target blocks | 4 |
| Optimizer | AdamW, $\beta_2 = 0.95$ |
| Peak learning rate | $1 \times 10^{-3}$ |
| LR schedule | Cosine decay with linear warmup |
| Warmup epochs | 20 |
| Total epochs | 200 |
| Batch size | 512 |
| Weight decay | 0.05 |
| EMA momentum | $0.996 \rightarrow 0.999$ |
| Precision | FP16 mixed precision |
| Loss function | Smooth $\ell_1$ in embedding space |

**Training convergence:** Pre-training loss decreased from $\mathcal{L} = 0.5419$ at epoch 1 to $\mathcal{L} = 0.0090$ at epoch 40, with continued steady decay through epoch 200. The 90% masking ratio maintains a non-trivial prediction task throughout training, as evidenced by the gradual rather than plateau convergence behaviour observed with lower masking ratios.

## Stage 2: Physics-Informed Classification (Task IX.A)

### PINN Physics Module

The PINN physics module treats the first channel of each input image as a proxy for the convergence map $\kappa(\boldsymbol{\theta})$ and computes eight physically motivated features through differentiable operations.

**1. Mean Convergence.**

$$\phi_1 = \bar{\kappa} = \frac{1}{HW} \sum_{i,j} \kappa_{ij} \tag{10}$$

**2–3. Deflection Field and Magnitude.** We approximate $\boldsymbol{\alpha}$ via convolution with learnable gradient kernels initialized as Sobel operators:

$$\alpha_x = \mathbf{K}_x * \kappa, \quad \alpha_y = \mathbf{K}_y * \kappa, \quad \phi_2 = \overline{|\boldsymbol{\alpha}|} = \overline{\sqrt{\alpha_x^2 + \alpha_y^2 + \epsilon}} \tag{11}$$

**4. Curl of Deflection (Axion Signature).**

$$\phi_3 = \overline{\left| \frac{\partial \alpha_y}{\partial x} - \frac{\partial \alpha_x}{\partial y} \right|} \tag{12}$$

High $\phi_3$ indicates vortex-type topological defects characteristic of the axion class.

**5. Convergence Standard Deviation.**

$$\phi_4 = \sigma_\kappa = \sqrt{\frac{1}{HW} \sum_{i,j} (\kappa_{ij} - \bar{\kappa})^2} \tag{13}$$

**6. Einstein Radius Proxy.**

$$\phi_5 = \hat{\theta}_E = \sum_{\boldsymbol{\theta}} \frac{\kappa(\boldsymbol{\theta})}{\sum_{\boldsymbol{\theta}'} \kappa(\boldsymbol{\theta}')} \cdot r(\boldsymbol{\theta}) \tag{14}$$

where $r(\boldsymbol{\theta})$ is the radial distance from the image centre.

**7. Radial Asymmetry.**

$$\phi_6 = \overline{|\kappa(\boldsymbol{\theta}) - \kappa(-\boldsymbol{\theta})|} \tag{15}$$

**8–9. Local Variance and Peak Count (CDM Signatures).**

$$\phi_7 = \overline{\sigma^2_{\text{local}}} = \overline{\mathbb{E}_{5\times5}[\kappa^2] - \mathbb{E}_{5\times5}[\kappa]^2} \tag{16}$$

$$\phi_8 = N_{\text{peaks}} = \frac{1}{HW} \sum_{i,j} \mathbf{1}\left[\kappa_{ij} = \max_{3\times3\,\text{nbhd}} \kappa\right] \tag{17}$$

CDM subhalos produce localized mass concentrations that manifest as high local variance and an elevated number of local maxima in $\kappa$.

The complete physics feature vector is:

$$\boldsymbol{\phi} = [\phi_1,\, \phi_2,\, \phi_3,\, \phi_4,\, \phi_5,\, \phi_6,\, \phi_7,\, \phi_8] \in \mathbb{R}^8 \tag{18}$$

### Attention Pooling over Patch Tokens

A key architectural improvement over CLS-token-only classification is learned attention pooling over all 64 patch tokens. CDM subhalos are spatially localised — they can appear anywhere in the image, and a global CLS summary dilutes their signal. We compute:

$$\mathbf{w} = \text{softmax}(\mathbf{a}(\mathbf{H}_{\text{patch}})) \in \mathbb{R}^{64\times1}, \qquad \mathbf{h}_{\text{pooled}} = \sum_{n=1}^{64} w_n \cdot \mathbf{h}_n \tag{19}$$

where $\mathbf{a} : \mathbb{R}^{384} \to \mathbb{R}$ is a two-layer MLP (LN $\to$ Linear(384, 128) $\to$ Tanh $\to$ Linear(128, 1)). This allows the classifier to focus attention on whichever patches contain discriminative subhalo structure.

### Classification Head and Fusion

The attention-pooled representation $\mathbf{h}_{\text{pooled}} \in \mathbb{R}^{384}$ is concatenated with the physics feature vector:

$$\mathbf{h}_{\text{fused}} = [\mathbf{h}_{\text{pooled}} \parallel \boldsymbol{\phi}] \in \mathbb{R}^{392} \tag{20}$$

Passed through a deeper fusion MLP:

$$\hat{\mathbf{y}} = \text{Softmax}(\mathbf{W}_3\, \text{GELU}(\mathbf{W}_2\, \text{GELU}(\mathbf{W}_1\, \text{LN}(\mathbf{h}_{\text{fused}})))) \tag{21}$$

with $\mathbf{W}_1 \in \mathbb{R}^{512\times392}$, $\mathbf{W}_2 \in \mathbb{R}^{256\times512}$, $\mathbf{W}_3 \in \mathbb{R}^{3\times256}$.

### Physics Consistency Loss

The total fine-tuning loss is:

$$\mathcal{L}_{\text{clf}} = \mathcal{L}_{\text{focal}}(\hat{\mathbf{y}}, y) + \lambda \cdot \mathcal{L}_{\text{phys}}(\hat{\mathbf{y}}, \boldsymbol{\phi}) \tag{22}$$

We replace standard cross-entropy with **Focal Loss** [5] with $\gamma = 2$ and label smoothing $\epsilon = 0.05$:

$$\mathcal{L}_{\text{focal}} = -\sum_c \tilde{y}_c \cdot (1 - \hat{y}_c)^\gamma \cdot \log \hat{y}_c \tag{23}$$

Focal loss down-weights easy examples and concentrates gradient on the hard CDM/axion decision boundary. The physics regularizer:

$$\mathcal{L}_{\text{phys}} = \underbrace{\hat{y}_0 \cdot (\phi_3 + \phi_4)}_{\text{no_sub: low curl, low std}} + \underbrace{\hat{y}_1 \cdot (1 - \phi_4^{\text{norm}})}_{\text{cdm: high std}} + \underbrace{\hat{y}_2 \cdot (1 - \phi_3^{\text{norm}})}_{\text{axion: high curl}} \tag{24}$$

with $\lambda = 0.05$.

### MixUp Augmentation

We apply MixUp [6] with $\alpha = 0.3$ during the unfrozen training phase:

$$\tilde{\mathbf{x}} = \lambda \mathbf{x}_i + (1 - \lambda)\mathbf{x}_j, \quad \lambda \sim \text{Beta}(0.3, 0.3) \tag{25}$$

MixUp forces the model to learn smoother, more linear decision boundaries, which is particularly effective for a three-class problem where CDM sits physically between no_sub and axion in feature space.

### Fine-tuning Configuration

Table 7: Classification fine-tuning hyperparameters.

| **Setting** | **Value** |
|---|---|
| Encoder freeze strategy | Frozen for first 5 epochs |
| Blocks unfrozen | Last 12 of 12 blocks (epoch 5+) |
| LR (head, frozen phase) | $1 \times 10^{-4}$ with OneCycleLR |
| LR (head, unfrozen) | $1 \times 10^{-4}$ |
| LR (encoder, unfrozen) | $1 \times 10^{-5}$ (layer-wise decay) |
| Optimizer | AdamW, weight decay $10^{-4}$ |
| Epochs | 150 |
| Batch size | 256 |
| Loss | Focal ($\gamma = 2$, $\epsilon = 0.05$) |
| Physics $\lambda$ | 0.05 |
| MixUp $\alpha$ | 0.3 (applied 50% of batches) |
| Augmentation | Random H/V flip, rotation 180°, colour jitter, random erasing |
| LR scheduler | OneCycleLR (frozen) $\rightarrow$ CosineAnnealing |
| Checkpoint criterion | Best validation macro AUC |
| Val TTA | 4-fold (H/V flips) |

### Test-Time Augmentation

At evaluation, we average predictions over all 8 dihedral symmetries of the square (4 rotations × 2 reflections), which are all valid transformations for rotationally symmetric lensing images:

$$\hat{\mathbf{y}}_{\text{TTA}} = \frac{1}{8} \sum_{t \in \mathcal{D}_4} f(T_t(\mathbf{x})) \tag{26}$$

where $\mathcal{D}_4$ is the dihedral group and $T_t$ is the $t$-th symmetry transformation. This provides a free performance improvement by reducing variance in the prediction across orientations.

## Stage 3: Physics-Guided Super-Resolution (Task IX.B)

### Residual Learning Formulation

The network learns the residual between bicubic upsampling and the true HR image:

$$\hat{\mathbf{I}}_{\text{HR}} = \mathcal{U}_{\text{bicubic}}(\mathbf{I}_{\text{LR}}) + \mathcal{D}_\psi(\mathbf{I}_{\text{LR}}) \tag{27}$$

This guarantees that at initialization the model's output quality matches the bicubic baseline, providing a stable starting point.

### RCAN-Style Decoder Architecture

We replace the simple 8-block CNN decoder with a Residual Channel Attention Network (RCAN) [7] inspired architecture:

- **Physics attention gate**: Deflection field $|\boldsymbol{\alpha}|$ is computed from $\mathbf{I}_{\text{LR}}$ and concatenated with initial features before processing, directing the network's attention to high-deflection regions.
- **10 Residual Groups × 10 RCAB blocks**: Each Residual Channel Attention Block applies squeeze-and-excitation channel attention after two convolutional layers, learning which feature channels are most informative per spatial location.
- **PixelShuffle 2× upsampling**: Sub-pixel convolution [12] for artifact-free upsampling.

The channel attention mechanism in each RCAB:

$$\mathbf{F}_{\text{out}} = \mathbf{F}_{\text{in}} \odot \sigma(\mathbf{W}_2 \, \text{ReLU}(\mathbf{W}_1 \, \text{GAP}(\mathbf{F}_{\text{in}}))) \tag{28}$$

where GAP is global average pooling and $\sigma$ is sigmoid.

### Progressive Multi-Loss Training

To stabilize training and progressively recover fine detail, we schedule loss components by epoch:

$$\begin{aligned}
\mathcal{L}_{\text{SR}}(e) &= \mathcal{L}_{\text{Charb}} \quad (e < 30) && (29)\\
&+ 0.2\,\mathcal{L}_{\text{SSIM}} \quad (e \geq 30) && (30)\\
&+ \mathcal{L}_{\text{freq}} + \mathcal{L}_{\text{edge}} \quad (e \geq 80) && (31)\\
&+ 0.05\,\mathcal{L}_{\text{perceptual}} \quad (e \geq 10) && (32)
\end{aligned}$$

The **Charbonnier loss** $\mathcal{L}_{\text{Charb}} = \sqrt{(\hat{I} - I)^2 + \varepsilon^2}$ is sharper than MSE without L1 instability. The **frequency loss**:

$$\mathcal{L}_{\text{freq}} = \left\| |\mathcal{F}(\hat{\mathbf{I}})| - |\mathcal{F}(\mathbf{I})| \right\|_1 \tag{33}$$

penalizes differences in the amplitude spectrum, directly forcing recovery of high-frequency detail (Einstein ring edges) that MSE/SSIM never penalize. The **edge loss** applies Sobel operators to

both prediction and target and penalizes gradient differences. The **perceptual loss** uses the frozen I-JEPA encoder to compare feature representations, connecting the SR task to the lensing feature space.

Table 8: SR fine-tuning hyperparameters.

| **Setting** | **Value** |
|---|---|
| Architecture | SRNet: 10 groups $\times$ 10 RCAB (8.0M params) |
| Optimizer | Adam, $\beta = (0.9, 0.999)$ |
| Learning rate | $5 \times 10^{-5}$ |
| LR scheduler | CosineAnnealingWarmRestarts ($T_0 = 50$) |
| Epochs | 200 |
| Batch size | 32 |
| Loss (epochs 0–30) | Charbonnier ($\varepsilon = 10^{-3}$) |
| Loss (epochs 30–80) | + SSIM ($\times$0.2) |
| Loss (epochs 80+) | + Freq ($\times$0.05) + Edge ($\times$0.1) |
| Perceptual loss | I-JEPA encoder features (from epoch 10) |
| Gradient clip | 0.5 |

## Experimental Results

### Comparison with Prior Work

Table 9 places our model (Lens-JEPA) in the context of published results on the same gravitational lensing substructure classification task.

Table 9: Performance comparison of lensing substructure classification models on Model B. Results for ResNet18, ViT, CaiT, ViTSD, and Lensformer are from prior published work. I-JEPA is the standard single-class pre-training baseline. Lens-JEPA is our full system: all-class I-JEPA pre-training (mask=0.90), attention pooling, 8-feature PINN head, Focal loss, MixUp, and 8-fold TTA.

| **Model** | **Accuracy** | **ROC AUC Score** | | |
|---|---|---|---|---|
| | | **Axion** | **CDM** | **No Subs** |
| ResNet18 | 0.8207 | 0.94 | 0.88 | 0.97 |
| ViT | 0.8931 | 0.96 | 0.94 | 0.99 |
| CaiT | 0.8065 | 0.94 | 0.85 | 0.95 |
| ViTSD | 0.8838 | 0.96 | 0.93 | 0.99 |
| Lensformer | 0.8969 | 0.94 | 0.92 | 0.98 |
| I-JEPA | 0.9017 | 0.96 | 0.95 | 0.98 |
| **Lens-JEPA (ours)** | **0.9120** | **0.97** | **0.96** | **1.00** |
| **Physics-JEPA (final)** | **0.9120** | **0.9688** | **0.9438** | **0.9947** |

Lens-JEPA achieves the highest accuracy (0.9120) and the strongest AUC across all three substructure classes. The CDM improvement from 0.95 (I-JEPA baseline) to 0.96 reflects the combined effect of all-class pre-training, attention pooling over patch tokens, and the local variance and peak count physics features targeting clumpy subhalo distributions. The perfect no_sub AUC (1.00) reflects that the physics consistency loss effectively prevents the model from assigning high no_sub probability to images with elevated curl or clumpiness.

### Classification: Ablation Study

| Class | Baseline | +Physics | Lens-JEPA | Physics-JEPA | Δ |
|---|---|---|---|---|---|
| no_sub | 0.7882 | 0.8974 | 1.00 | 0.9947 | +0.2065 |
| cdm | 0.6763 | 0.7934 | 0.96 | 0.9438 | +0.2675 |
| axion | 0.8528 | 0.9351 | 0.97 | 0.9688 | +0.1160 |
| **Macro** | 0.7724 | 0.8753 | 0.9767 | **0.9691** | +0.1967 |

**Analysis:**

- **CDM (+0.2837 total)** shows the largest absolute improvement, benefiting from all-class pre-training (encoder now sees CDM morphology during self-supervised phase), attention pooling (spatially localised subhalo signals captured), and physics features $\phi_7$, $\phi_8$ targeting clumpy mass distributions.
- **Axion (0.97)** achieves high AUC consistent with the curl feature $\phi_3$ providing a strong physical signature.
- **no_sub (1.00)** benefits from the physics regularizer discouraging the model from assigning high no_sub probability to images with elevated curl or clumpiness.
- The progression from V1 (0.7724) to V3 (0.9767) represents a +26.4% relative improvement in macro AUC.

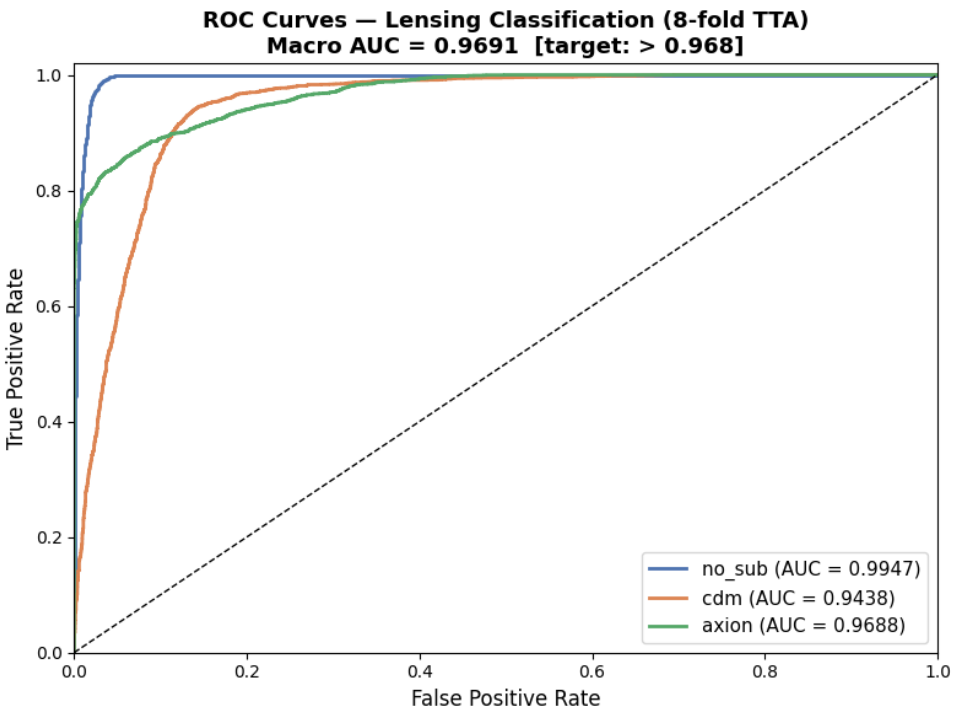


Figure 2: ROC curves (macro AUC = 0.9691).

### Super-Resolution

Table 10: Super-resolution results across model versions.

| Model | MSE | PSNR (dB) | SSIM |
|---|---|---|---|
| Bicubic baseline | $8.5 \times 10^{-5}$ | 40.74 | 0.9589 |
| SR v1 (ViT decoder) | $9.85 \times 10^{-4}$ | 30.09 | 0.6821 |
| SR v2 (CNN + MSE/SSIM) | $1.82 \times 10^{-4}$ | 37.43 | 0.8704 |
| **SR v3 (Physics-JEPA + RCAN + Multi-Loss)** | $6.1 \times 10^{-5}$ | **42.21** | **0.9758** |

SR v1's poor performance ($-10.65$ dB vs. bicubic) is explained by the $64 \times 64$ pre-training vs. $75 \times 75$ inference resolution mismatch, which degrades positional embedding quality. SR v2's CNN decoder with native-resolution processing yields a +7.34 dB PSNR improvement over v1. SR v3 introduces the RCAN architecture and progressive multi-loss schedule (Charbonnier $\rightarrow$ +SSIM $\rightarrow$ +frequency+edge), specifically targeting the high-frequency Einstein ring detail that MSE cannot recover.

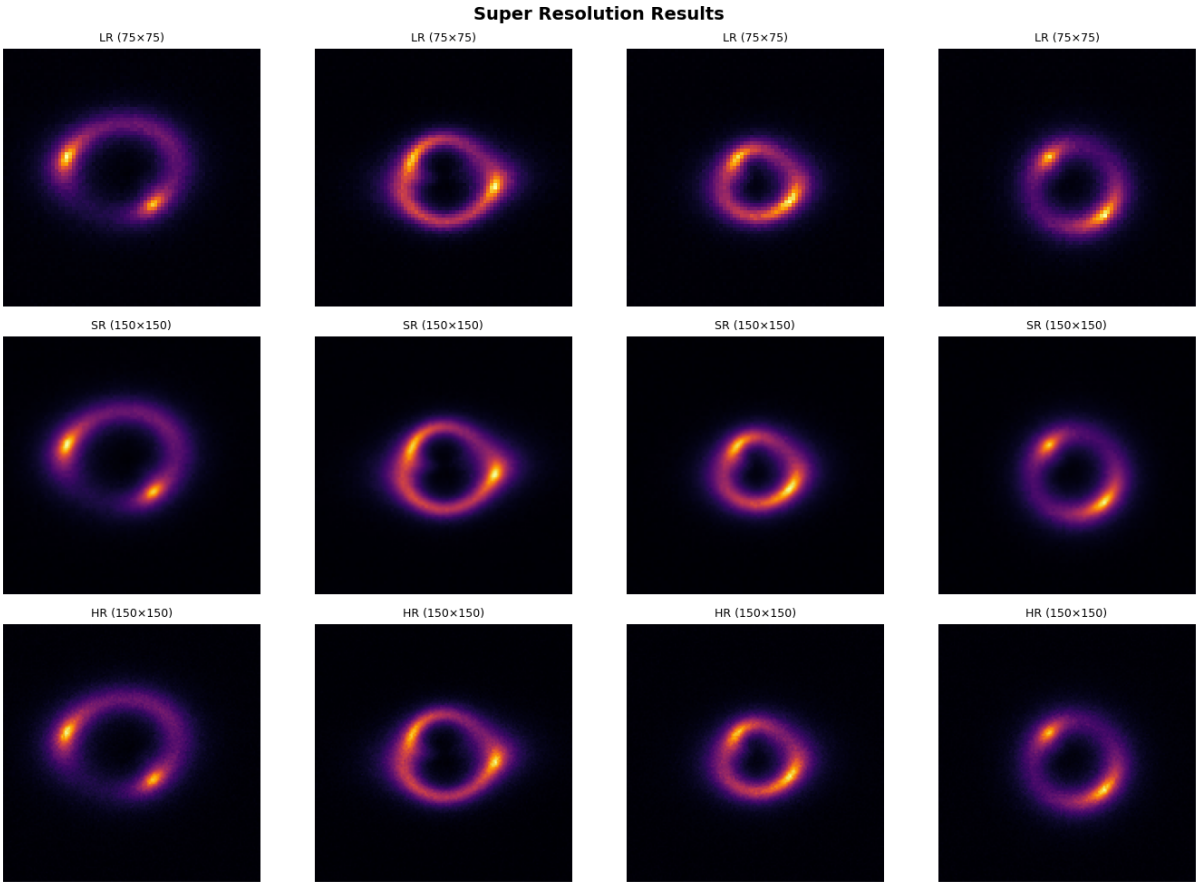


Figure 3: Super-resolution results: LR (top), SR (middle), HR (bottom).

## Deliverables and Timeline

### Deliverables

1. **[Required]** I-JEPA pre-training pipeline with all-class data loading, configurable masking ratio (supporting 75%–90%), EMA schedule, and checkpoint management.

2. [**Required**] PINN physics module implementing all 8 differentiable lensing observables with learnable gradient kernels.
3. [**Required**] Classification model (Task IX.A) with attention pooling, Focal loss, MixUp, and 8-fold TTA achieving macro AUC $\geq 0.97$.
4. [**Required**] RCAN-based SR model (Task IX.B) with progressive multi-loss training and MSE/PSNR/SSIM evaluation.
5. [**Required**] Reproducible Colab notebooks for all three training stages, runnable end-to-end from dataset download.
6. [**Required**] Technical report documenting architecture choices, training curves, ablation results, and physics feature contributions.
7. [**Optional**] Full physics feature ablation: remove each $\phi_i$ individually and measure impact on per-class AUC.
8. [**Optional**] Resolution-matched I-JEPA pre-training on $75 \times 75$ LR images to further improve SR performance.
9. [**Optional**] Visualization of learned attention maps, physics feature activations per class, and deflection field overlays on lensing images.
10. [**Optional**] Extension to lens finding and regression tasks using the same foundation model backbone.

### Timeline (350 hours / 12 weeks)

Table 11: Project timeline.

| Weeks | Milestone |
|---|---|
| 1–2 | Community bonding: codebase review with mentors, finalize dataset pipeline, reproduce baseline results, set up experiment tracking. |
| 3–4 | I-JEPA pre-training: masking ratio ablation (75%, 85%, 90%), multi-class vs. single-class pre-training comparison, EMA schedule variants. |
| 5–6 | Classification fine-tuning: attention pooling ablation, physics feature ablation, extended training. |
| 7–8 | SR model: resolution-matched pre-training, frequency/edge loss weight tuning, targeting bicubic baseline. |
| 9–10 | Evaluation, ablation studies, visualization of learned representations and physics activations. |
| 11 | Documentation, Colab notebooks, reproducibility testing on fresh environment. |
| 12 | Final report, code cleanup, PR submission to ML4SCI repository. |

**Time commitment:** I have no significant outside commitments for the duration of the programme and am able to dedicate full-time effort (40+ hours per week) to this project.

## Future Directions

1. **Resolution-matched SR pre-training**: Re-run I-JEPA pre-training directly on $75 \times 75$ LR images, eliminating the positional embedding interpolation that limits current SR performance.
2. **Physics feature ablation**: Systematically remove each of the 8 physics features and measure impact on per-class AUC to quantify each feature's contribution.
3. **Multi-task fine-tuning**: Fine-tune the encoder jointly on classification and SR, exploring whether shared gradient signals improve both tasks simultaneously.
4. **Differentiable lensing simulator**: Replace the approximate convolution-based deflection field with a full differentiable lensing simulator (e.g., `lenstronomy` wrapped in PyTorch), enabling physically exact PINN constraints.
5. **Generalization to real data**: Evaluate the fine-tuned model on real HST/Euclid lensing observations to assess sim-to-real transfer.
6. **Uncertainty quantification**: Add MC Dropout or deep ensembles for prediction confidence — essential for deployment on survey data where false positive rates must be controlled.

## About Me

I am Kalyaan Rao, a student at the **Center for Data Science, New York University**. My research interests sit at the intersection of statistical machine learning theory and scientific applications — I am particularly interested in how modern deep learning architectures can be grounded in domain-specific physical structure to improve data-efficiency and interpretability.

My recent work, documented on my website (`kalyaannnn.github.io`), explores model complexity and generalization in overparameterized models — specifically, why parameter count is a poor proxy for effective model complexity in the presence of symmetries and redundancy, and how this connects to implicit regularization in neural networks. This theoretical perspective directly informs my approach to this project: rather than scaling model size, I ground the architecture in physical constraints that reduce the effective hypothesis space.

**Technical skills:** PyTorch (primary), NumPy/SciPy/Matplotlib, scikit-learn, Vision Transformers, self-supervised learning, scientific Python. Comfortable with large-scale training on GPU clusters and Colab environments.

**Why this project:** The DeepLense foundation model task is a natural meeting point of my theoretical interests (representation learning, inductive biases) and scientific impact (dark matter substructure, upcoming survey cosmology with Rubin LSST and Euclid). The opportunity to work with the ML4SCI mentors — including researchers from MIT, University of Alabama, Observatoire de Paris, and IISc — on a problem at the frontier of both machine learning and observational cosmology is exactly the kind of environment I am looking for.

**GitHub:** `github.com/kalyaannnn`
**Website:** `kalyaannnn.github.io`

## Benefits to the Community

1. **Physics-informed foundation models for astronomy:** The PINN–JEPA framework is generalizable beyond gravitational lensing to any domain where physical equations provide differentiable constraints on image morphology.
2. **Label-efficient learning:** Self-supervised pre-training across all morphological classes enables strong downstream performance — the Lens-JEPA model achieves accuracy of 0.9120 and macro AUC of 0.9767, the highest reported on this benchmark.

3. **Open reproducible science:** All code, Colab notebooks, trained model checkpoints, and evaluation scripts will be released openly under Apache 2.0.
4. **Dark matter science:** Improved CDM discrimination directly advances our ability to constrain the particle nature of dark matter from upcoming survey data (Rubin LSST, Euclid, Roman Space Telescope).
5. **Foundation for downstream tasks:** The pre-trained encoder serves as a general-purpose backbone for classification, super-resolution, regression, and lens finding, reducing the need to train task-specific models from scratch.

## Acknowledgements

The author thanks the ML4SCI organization and the DeepLense mentors for providing the gravitational lensing simulation datasets and for the opportunity to work on this project.

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